

22.10.2024

EXPERIMENT-8

Credit Risk Assessment (Decision Tree)

```
1 # Import necessary libraries
2 import pandas as pd
3 from sklearn.model_selection import train_test_split
4 from sklearn.tree import DecisionTreeClassifier
5 from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, classification_report
6 from sklearn.impute import SimpleImputer
7 from sklearn.preprocessing import LabelEncoder
```

```
1 # Load the dataset
2 data = pd.read_csv('Loan_Default.csv')
```

```
1 # Data Preprocessing
2 print(data.head())
3 print(data.info())
```

```
   ID  year  loan_limit  Gender  approv_in_adv  loan_type  \
0  24890  2019         cf  Sex Not Available      nopre    type1
1  24891  2019         cf      Male      nopre    type2
2  24892  2019         cf      Male       pre    type1
3  24893  2019         cf      Male      nopre    type1
4  24894  2019         cf     Joint       pre    type1

   loan_purpose  Credit_Worthiness  open_credit  business_or_commercial  ...  \
0          p1                l1        nopc          nob/c  ...
1          p1                l1        nopc           b/c  ...
2          p1                l1        nopc          nob/c  ...
3          p4                l1        nopc          nob/c  ...
4          p1                l1        nopc          nob/c  ...

   credit_type  Credit_Score  co-applicant_credit_type  age  \
0          EXP           758                CIB  25-34
1          EQUI           552                EXP  55-64
2          EXP           834                CIB  35-44
3          EXP           587                CIB  45-54
4          CRIF           602                EXP  25-34

   submission_of_application  LTV  Region  Security_Type  Status  dtir1
0          to_inst  98.728814  south    direct         1  45.0
1          to_inst      NaN  North    direct         1   NaN
2          to_inst  80.019685  south    direct         0  46.0
3          not_inst  69.376900  North    direct         0  42.0
4          not_inst  91.886544  North    direct         0  39.0
```

```
[5 rows x 34 columns]
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 148670 entries, 0 to 148669
Data columns (total 34 columns):
```

#	Column	Non-Null Count	Dtype
0	ID	148670 non-null	int64
1	year	148670 non-null	int64
2	loan_limit	145326 non-null	object
3	Gender	148670 non-null	object
4	approv_in_adv	147762 non-null	object
5	loan_type	148670 non-null	object
6	loan_purpose	148536 non-null	object
7	Credit_Worthiness	148670 non-null	object
8	open_credit	148670 non-null	object
9	business_or_commercial	148670 non-null	object
10	loan_amount	148670 non-null	int64
11	rate_of_interest	112231 non-null	float64
12	Interest_rate_spread	112031 non-null	float64
13	Upfront_charges	109028 non-null	float64
14	term	148629 non-null	float64
15	Neg_ammortization	148549 non-null	object
16	interest_only	148670 non-null	object
17	lump_sum_payment	148670 non-null	object
18	property_value	133572 non-null	float64
19	construction_type	148670 non-null	object
20	occupancy_type	148670 non-null	object
21	Secured_by	148670 non-null	object
22	total_units	148670 non-null	object
23	income	139520 non-null	float64
24	credit_type	148670 non-null	object
25	Credit_Score	148670 non-null	int64
26	co-applicant_credit_type	148670 non-null	object
27	age	148470 non-null	object
28	submission_of_application	148470 non-null	object
29	LTV	133572 non-null	float64
30	Region	148670 non-null	object
31	Security_Type	148670 non-null	object
32	Status	148670 non-null	int64
33	dtir1	124549 non-null	float64

dtypes: float64(8), int64(5), object(21)

memory usage: 38.6+ MB

None

```

1 # Handle missing values
2 # For numeric columns, use the mean strategy; for categorical columns, use the most frequent strategy
3 numeric_cols = data.select_dtypes(include=['float64', 'int64']).columns
4 categorical_cols = data.select_dtypes(include=['object']).columns
5
6 # Impute missing values
7 imputer_numeric = SimpleImputer(strategy='mean')
8 imputer_categorical = SimpleImputer(strategy='most_frequent')
9
10 data[numeric_cols] = imputer_numeric.fit_transform(data[numeric_cols])
11 data[categorical_cols] = imputer_categorical.fit_transform(data[categorical_cols])

```

```

1 # Feature engineering - Encode categorical variables
2 label_encoders = {}
3 for col in categorical_cols:
4     le = LabelEncoder()
5     data[col] = le.fit_transform(data[col])
6     label_encoders[col] = le

```

```

1 # Split the dataset into features (X) and target (y)
2 # Replace 'Status' with the actual column that represents loan status
3 X = data.drop('Status', axis=1) # Features
4 y = data['Status'] # Target
5
6 # Train-Test Split
7 X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

```

```

1 # Build Decision Tree Model
2 dt_model = DecisionTreeClassifier(random_state=42)
3 dt_model.fit(X_train, y_train)
4
5 # Make Predictions
6 y_pred = dt_model.predict(X_test)

```

```

1 # Evaluate the Model
2 accuracy = accuracy_score(y_test, y_pred)
3 precision = precision_score(y_test, y_pred, average='weighted')
4 recall = recall_score(y_test, y_pred, average='weighted')
5 f1 = f1_score(y_test, y_pred, average='weighted')
6
7 # Print evaluation metrics
8 print(f"Accuracy: {accuracy}")
9 print(f"Precision: {precision}")
10 print(f"Recall: {recall}")
11 print(f"F1 Score: {f1}")
12 print("Classification Report:\n", classification_report(y_test, y_pred))

```

Accuracy: 0.9998318423353737

Precision: 0.9998318532787672

Recall: 0.9998318423353737

F1 Score: 0.9998318462725634

Classification Report:

	precision	recall	f1-score	support
0.0	1.00	1.00	1.00	22494
1.0	1.00	1.00	1.00	7240
accuracy			1.00	29734
macro avg	1.00	1.00	1.00	29734
weighted avg	1.00	1.00	1.00	29734

```

1 # Feature Importance
2 importances = dt_model.feature_importances_
3 feature_names = X.columns
4 important_features = pd.DataFrame({'Feature': feature_names, 'Importance': importances}).sort_values(by='Importance', aso
5
6 # Print important features
7 print(important_features)

```

	Feature	Importance
11	rate_of_interest	9.931370e-01
12	Interest_rate_spread	2.909962e-03
24	credit_type	2.511306e-03
15	Neg_ammortization	8.260225e-04
23	income	2.390899e-04
2	loan_limit	1.370421e-04
18	property_value	8.006712e-05
9	business_or_commercial	5.950465e-05
25	Credit_Score	4.821567e-05
13	Upfront_charges	3.958360e-05
22	total_units	1.139481e-05
3	Gender	8.288847e-07
0	ID	0.000000e+00
21	Secured_by	0.000000e+00
27	age	0.000000e+00
28	submission_of_application	0.000000e+00
29	LTV	0.000000e+00
30	Region	0.000000e+00
31	Security_Type	0.000000e+00
26	co-applicant_credit_type	0.000000e+00
16	interest_only	0.000000e+00
20	occupancy_type	0.000000e+00
19	construction_type	0.000000e+00
17	lump_sum_payment	0.000000e+00
1	year	0.000000e+00
14	term	0.000000e+00
10	loan_amount	0.000000e+00
8	open_credit	0.000000e+00
7	Credit_Worthiness	0.000000e+00
6	loan_purpose	0.000000e+00
5	loan_type	0.000000e+00
4	approv_in_adv	0.000000e+00
32	dtir1	0.000000e+00